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Problem

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In this practice problem you will derive a differential equation and complete a first modelling cycle. But first, the problem!



A daycare center for two- to four-year-olds has acquired a new outside playground with grass, plants and a sandbox. Summer is coming and everyone is looking forward to let the children play outside. However, then the staff starts to worry about sun protection. Can they let the children play outside in the middle of the day? Is the sun screen they use sufficient?

First, information is collected:

- Sunburn is considered a long term health risk, especially in small children.
- In this country, without protection, lighter skinned children can be in the sun on a summers day between noon and 3 PM for only 10 minutes before their skin reddens and starts to get sunburned. Darker skinned children can stay out somewhat longer (20 to 30 minutes), but also not indefinitely before they develop sunburn as well.
- The daycare center uses sunscreen lotion with a Sun Protection Factor (SPF) of 30. This means that if the cream is applied in a thickness of 2 mg per square cm skin, you can stay out in the sun 30 times as long as without sunscreen.

- Most people apply a much thinner sunscreen layer than 2 mg per square cm skin. Most people apply the suncream unevenly, and with an *average* thickness of only 0.5 mg per square cm.

The daycare center instructs the staff to apply the suncream carefully, generously and often, on all exposed skin of all the children when they play outside. But, to be on the safe side, they estimate that their layers are only 0.5 mg per square cm skin.

They first want to know how this thinner layer affects the Sun Protection Factor of their suncream. And then, they want estimates for how long the children can safely play outside midday on a summers day.

Exercise A

1/1 point (graded)

Start with defining the light quantity. We only need relative light quantities: everything can be defined relative to the UV light of the sun between noon and 3 PM on a cloudless summers day, that causes sunburn in unprotected light skinned people after 10 minutes. Call this light intensity: I_{full} .

(Its units are not important, but could be the energy of the UV part of the sunlight that falls in a second on a horizontal square meter.)

If a layer of sun screen lotion with a SPF=30 is applied, then the intensity of the UV light that reaches the skin equals a factor α times I_{full} . What is α ?

✓ Answer: 1/30

Submit

You have used 1 of 3 attempts

i Answers are displayed within the problem

Exercise B

1/1 point (graded)

The suncream lotion layer is calibrated for 2 mg per cm^2 . Assume that the density of the lotion is that of water: 1 g per cm^3 .

What is the thickness of this layer of sunscreen in micrometer? ($1 \mu\text{m} = 10^{-4} \text{ cm}$.) (For comparison: a human hair has a diameter of 17 to 181 μm .)

20

✓ Answer: 20

20

Submit

You have used 2 of 3 attempts

i Answers are displayed within the problem

Now that the sunscreen problem and its relevant quantities have been defined, in the next section you will derive a mathematical model.

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